



Two-Stage Ultrafiltration for Enzyme-Product Separation

Bioprocess Enzyme Recovery & Product Purification · Gropello, Italy

SITE INFORMATION

- LOCATION
- Gropello, Italy
- INDUSTRY
- Pharmaceutical & Biotech
- SYSTEM SUPPLIED
- UF (Two Stage)
- CAPACITY DESIGNED
- 5,000 litres/hr
- CLIENT
- European Pharmaceutical Major
- 100% of all three oxidative enzymes separated from reaction mixture
 - >98% product recovery in UF permeate
 - Permeate shows Nil enzyme activity
 - CE-certified, hot-water sanitizable design

SCOPE OF SUPPLY

Two stage fully automatic PLC-based system for Process and CIP cycles.

OBJECTIVE

Separation of oxidative enzymes from product post bio-transformation, with complete recovery of both product and enzymes.

DESIGN FEATURES

Sanitary design with hot water sanitizable, customized MWCO UF membranes to maximize process separation. System built under CE guidelines and certified by a CE auditor. Diafiltration provision built into the design for enzyme recycle.

100% Enzyme Separation · >98% Product Recovery · Nil Permeate Enzyme Activity · CE-Certified

INTRODUCTION

Our client, a European Pharmaceutical major, had developed a bioprocess in place of a conventional chemical process for synthesizing one of their products. The process uses an enzyme for Enzymatic Regioselective Oxidation, a cleaner route offering better yield and purity than the conventional chemical alternative.

Post enzymatic conversion of substrate to product, it was critical not only to separate the enzyme from the reaction mixture, but also to recover the maximum amount of product to send forward to the next processing step.



Permionics two-stage UF system installed at the client's production facility in Italy.

THE CHALLENGE

The Problem. The client needed to separate three distinct oxidative enzymes from the product stream following bio-transformation, while recovering as much of the high-value product as possible for the next conversion step. Both the enzymes and the product carried significant value, so the separation needed to capture both streams cleanly rather than treat one as waste.

Path to Production. The client first invested in a Permionics laboratory pilot system at their own R&D laboratory, where several ultrafiltration membranes were screened and trialed on lab-scale batch volumes to establish the correct membrane type and design parameters for the scaled-up process. Only after successfully establishing the ultrafiltration process at laboratory scale was the full process — from oxidation through enzyme recovery and reduction — scaled up in their production facility in Italy.

THE TECHNICAL SOLUTION

System Design. Permionics supplied a two-stage ultrafiltration system, fully automatic and PLC-based for both process and CIP (Clean-in-Place) cycles, designed for a capacity of 5,000 litres/hr.

Sanitary & Regulatory Design. The system uses hot water sanitizable, customized MWCO (molecular weight cut-off) UF membranes selected to maximize process separation. The complete system was built under CE guidelines and certified by a CE auditor, meeting the regulatory requirements of a European pharmaceutical production environment.

Recovery & Recycle Provision. The design incorporates provision for diafiltration of the concentrated enzymes in the final reject stream, allowing all recovered enzyme to be purified and recycled back into the next reaction with only a minor top-up required.

OPERATIONAL RESULTS

100% OXIDATIVE ENZYMES SEPARATED FROM MIXTURE	>98% PRODUCT RECOVERY IN UF PERMEATE	Nil ENZYME ACTIVITY IN PERMEATE	5,000 LITRES/HR CAPACITY DESIGNED
Performance Parameter		Outcome	
Oxidative enzyme separation from reaction mixture		100% — Nil activity for all three enzymes in permeate	
Product recovery (free from enzymes)		>98% in UF permeate	
Enzyme recovery from final reject		Concentrated, with diafiltration provision built in for recycle with minor top-up	
Regulatory compliance		Built under CE guidelines, certified by CE auditor	
Membrane design		Hot water sanitizable, customized MWCO UF membranes	

CONCLUSIONS

Permionics' two-stage ultrafiltration system enabled this European pharmaceutical major to commercially scale a cleaner, enzyme-based bioprocess in place of their conventional chemical synthesis route. At commercial production scale in Italy, the system separates 100% of all three oxidative enzymes from the reaction mixture, with permeate showing nil enzyme activity, while simultaneously recovering more than 98% of the product free from enzymes for the next process step. The system delivers dual value from a single unit operation — purifying the product in the permeate while concentrating and preserving the enzymes in the reject for recycle via diafiltration — all within a sanitary, CE-certified design built for hot water sanitization and long-term reliability in a regulated pharmaceutical environment.